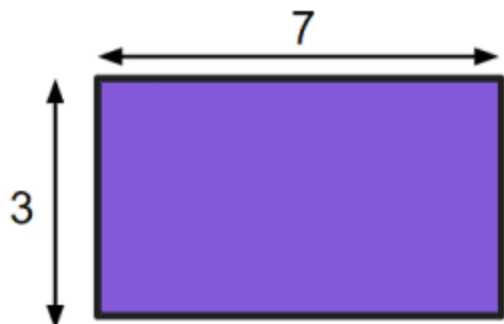




Problem solving with expressions and formulae

- 1 All the lengths in the diagram are given in centimetres. Which calculation will find the area of the rectangle? Tick **1** correct answer

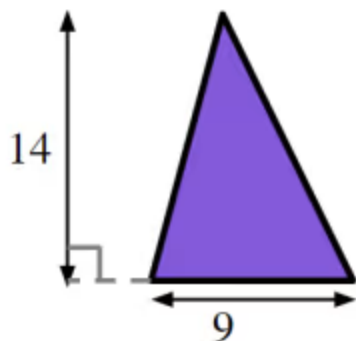


- ☐ $7 + 3$
☐ $\frac{7}{3}$
☒ 7×3
☐ $2(7 + 3)$
☐ $2 \times 7 + 2 \times 3$

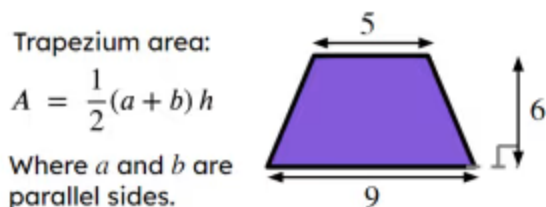
- 2 Select the expression that is equivalent to $\frac{1}{2}(6x - 2)$. Tick **1** correct answer

- ☒ $3x - 1$
☐ $3x - 2$
☐ $6x - 1$
☐ $12x - 2$
☐ $12x - 4$

- 3 All the lengths in the diagram are given in centimetres. The area of the triangle is 63 cm^2 . Fill in the blank



- 4 All the lengths in the diagram are given in metres. The area of the trapezium is 42 m². Fill in the blank



- 5 Match each product of two binomials to its expanded form. Write the correct letter in each box

a	$(x + 5)(x + 1)$
b	$(x + 5)(x - 1)$
c	$(x - 5)(x + 1)$
d	$(x - 5)(x - 1)$
e	$(x + 2)^2$
f	$(x - 2)^2$

b	$x^2 + 4x - 5$
f	$x^2 - 4x + 4$
d	$x^2 - 6x + 5$
e	$x^2 + 4x + 4$
a	$x^2 + 6x + 5$
c	$x^2 - 4x - 5$

- 6 Which of these is the correct expansion of $(4x + 3)^2$? Tick **1** correct answer

- ☐ $8x^2 + 12x + 9$
☐ $8x^2 + 14x + 9$
☐ $16x^2 + 9$
☐ $16x^2 + 6x + 9$
☒ $16x^2 + 24x + 9$

